Q4 (c)(

To show U is normal in T

we will show $\tau U \tau^{-1} \subset U$ for all $\tau \in T$,

So take any $\tau = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$, $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \in U$

and consider.

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \frac{1}{ac} \begin{pmatrix} c & -b \\ 0 & a \end{pmatrix}$$

$$= \frac{1}{ac} \begin{pmatrix} a & ax+b \\ 0 & c \end{pmatrix} \begin{pmatrix} c & -b \\ 0 & a \end{pmatrix}.$$

$$= \frac{1}{ac} \begin{pmatrix} ac & -ab + a^2x + ab \\ 0 & ac \end{pmatrix}$$

$$= \frac{1}{ac} \begin{pmatrix} ac & a^2x \\ 0 & ac \end{pmatrix}$$

$$= \begin{pmatrix} 1 & a^2x/ac \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \frac{ax}{c} \\ 0 & 1 \end{pmatrix} \in U.$$

Therefore $\forall \tau \in T \quad \tau U \tau^{-1} \subset U$.

So U is normal in T by theorem 10.3.

d). Consider the factor group T/U.

T/U is abelian means

$$\forall \alpha, \beta \in T \qquad \alpha U \beta U = \beta U \alpha U$$

$$\iff \quad (\alpha\beta) U = (\beta\alpha) U.$$

From Lemma 6.3 (5)

$$\iff \quad \underline{\underline{(\alpha\beta)^{-1} \beta\alpha \in U}}$$

check this.

Let $\alpha, \beta \in T$

Let's say $\alpha = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$, $\beta = \begin{pmatrix} d & e \\ 0 & f \end{pmatrix}$

(a, c, d, f all non-zero).

Consider

$$(\alpha\beta)^{-1} \beta\alpha = \beta^{-1} \alpha^{-1} \beta\alpha$$

$$= \begin{pmatrix} d & e \\ 0 & f \end{pmatrix}^{-1} \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}^{-1} \begin{pmatrix} d & e \\ 0 & f \end{pmatrix} \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$$

$$= \frac{1}{ac}\frac{1}{df}\begin{pmatrix} f & -e \\ 0 & d \end{pmatrix}\begin{pmatrix} c & -b \\ 0 & a \end{pmatrix}\begin{pmatrix} d & e \\ 0 & f \end{pmatrix}\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$$

$$= \frac{1}{ac}\frac{1}{df}\begin{pmatrix} fc & -bf-ae \\ 0 & ad \end{pmatrix}\begin{pmatrix} da & db+ec \\ 0 & fc \end{pmatrix}$$

$$= \frac{1}{acdf}\begin{pmatrix} adcf & \sim\sim \\ 0 & adfc \end{pmatrix}$$

$$= \begin{pmatrix} 1 & \sim\sim \\ 0 & 1 \end{pmatrix} \in U$$

(e). Is T normal in $GL_2(\mathbb{R})$.

From theorem 3 we shall check that

$\forall\, \alpha \in GL_2(\mathbb{R})$ $\alpha T \alpha^{-1} \subset T$.

Let's say $\alpha = \begin{pmatrix} d & e \\ f & g \end{pmatrix} \in GL_2(\mathbb{R})$

where $dg - ef \neq 0$.

and consider

$$\begin{pmatrix} d & e \\ f & g \end{pmatrix} \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} d & e \\ f & g \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} 3/2 & -1/2 \\ 1/2 & 1/2 \end{pmatrix} \notin T.$$

when $a = \dots = f = 1$, $g = -1$.

Therefore T is <u>not</u> normal in G.

<u>Q5</u> Let G be a group with normal subgroups N, M.

From chapter we know $N \cap M$ is a subgroup of G. From theorem 10.3 we will show $\boxed{\forall g \in G. \quad g(N \cap M) g^{-1} \subset N \cap M.}$

where we can use the fact that

$$\boxed{\forall g \in G \quad \underset{①}{g N g^{-1} \subset N}, \quad \underset{②}{g M g^{-1} \subset M}}$$

Let $g \in G$ and let $x \in N \cap M$.

Consider gxg^{-1}

since $x \in N$, $gxg^{-1} \in N$. from ①

since $x \in M$, $gxg^{-1} \in M$ from ②

Therefore $gxg^{-1} \in N \cap M$.

Therefore $\forall g \in G$ $g(N \cap M) g^{-1} \subset N \cap M$.

<u>Q6</u> Assume G is abelian

and consider a factor G/H.

with a normal subgroup H of G.

Let $aH, bH \in G/H$.

$$(aH)(bH) = (ab)H, \quad \text{def of op in } G/H.$$

$$= (ba)H, \quad \text{since } G \text{ is abelian.}$$

$$\downarrow$$

$$= (bH)(aH), \quad \text{def of op in } G/H.$$

Therefore G/H is abelian.

Q8. Assume G is cyclic.

i.e. $\exists\ \underline{\underline{g}} \in G$ where $G = \langle g \rangle$
$= \{ g^k : k \in \mathbb{Z} \}$.

Consider the factor group G/H and a typical element aH.

We need to show that there is a generator for G/H. We propose that it is the coset gH.

Observe that

$$aH = (g^k)H \underset{\text{by the gp op. def in } G/H.}{=} (gH)^k$$

for some $k \in \mathbb{Z}$ since $G = \langle g \rangle$ ~~for~~ $k \in \mathbb{Z}$.

Therefore G/H is cyclic

Q9. Prove ~~and~~ or disprove.

$$\boxed{\text{If } H \text{ and } G/H \text{ are cyclic, then } G \text{ is cyclic.}}$$

Let's attempt and proof:

Assume H and G/H are cyclic.

ie $H = \langle h \rangle$ for some $h \in H$

and $G/H = \langle aH \rangle$ for some $a \in G$.

Let's propose the generator $\underline{\quad a \quad}$ for G.

Let $x \in G$

Observe that

$$(xH) = (aH)^k = a^k H.$$

$$\Rightarrow x^{-1} a^k \in H.$$

$$\Rightarrow x^{-1} a^k = h^l, \text{ for some } l \in \mathbb{Z}.$$

$$\Rightarrow x = a^k h^{-l}$$

Looks tricky!

$$\overset{?}{=} \underline{\quad\quad}, \text{ for some } \cancel{k} \in \mathbb{Z}.$$

Therefore G is cyclic.

So maybe the result is false, so can produce a counter example contradicting ↑

a non-cyclic G with cyclic normal subgroup H and cyclic factor group G/H.

Consider D_4, not cyclic.

It has a normal subgroup

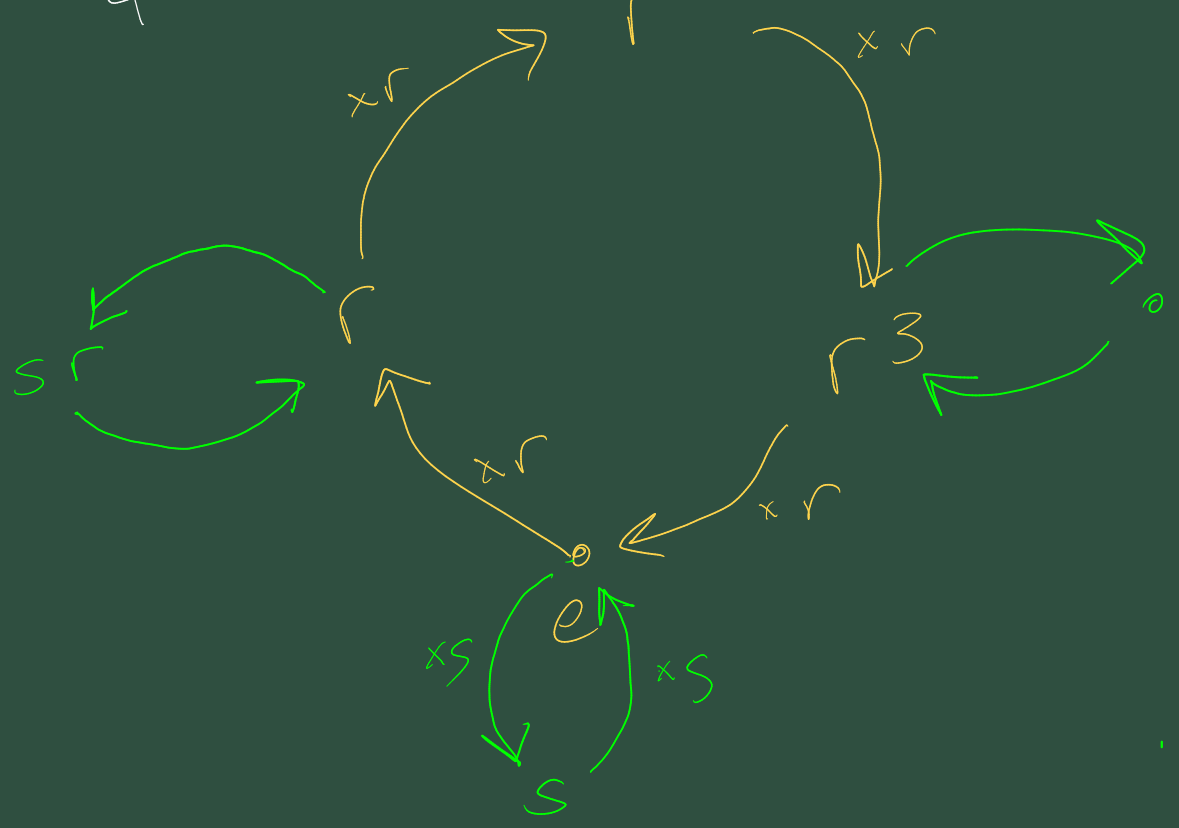
$$R_1 = \langle r \rangle = \{e, r, r^2, r^3\}.$$

and the factor group was

$$D_4/R_1 \cong \mathbb{Z}_2, \text{ which is cyclic.}$$

$$= \{0, 1\} = \langle 1 \rangle.$$

D_4



Remember

$$(aH)(bH) = (ab)H.$$

$$(aH)^2 = (a^2)H$$

$$(aH)^3 = (a^3)H.$$